JIA, Mengzhao (she/her)

355 Fitzpatrick Hall, Notre Dame, IN 46556
jiamengzhao98@gmail.com Google Scholar Personal Website

Research Interest: Reasoning. Multimodal Large Language Models. Agentic Physical AI.

Education

Ph.D. in Computer Science and Engineering 08/2023 – 07/2027 (expected)
University of Notre Dame, USA. Advisor: Prof. Meng Jiang

M. S. in Computer Science and Engineering 09/2020 – 06/2023
Shandong University, China. Advisor: Prof. Liqiang Nie

B. Eng in Electronic Science and Technology 09/2016 – 06/2020
Shandong University, China.

Work Experience

Research Intern at Orby AI, Mountain View, US. *Worked on Pub.[2]* 03/2025 – 09/2025

Research Intern at Tencent AI Lab, Seattle, US. *Worked on Pub.[4]* 05/2024 – 09/2024

Recent Work

1. **Mengzhao Jia**, Yang Lin, Xixin Zhang, Zhihan Zhang, Xiaobai Liu, Meng Jiang. *Agent as Policy for Robotic Manipulation.* arXiv 2026. [arXiv] [Code] [Project Page] [HuggingFace]
   AGP uses a general purpose agent as a robot policy, combining visual reasoning, programming, and physical feedback to perform zero-shot manipulation tasks.
2. **Mengzhao Jia**, Zhihan Zhang, Ignacio Cases, Zheyuan Liu, Meng Jiang, Peng Qi. *AutoRubric: Rubric-Based Generative Rewards for Faithful Multimodal Reasoning.* Findings of **ACL** 2026 [arXiv] [Code]
   AutoRubric uses automatically generated rubrics in multimodal reinforcement learning to improve both accuracy and reasoning faithfulness.
3. **Mengzhao Jia**, Zhihan Zhang, Meng Jiang. *Prioritizing the Best: Incentivizing Reliable Multimodal Reasoning by Rewarding Beyond Answer Correctness.* Findings of **EMNLP** 2026. [arXiv]
   We use an LLM judge to compare the relative reasoning quality of trajectories for the same question and assign rewards accordingly, improving multimodal reasoning reliability through reinforcement learning.
4. **Mengzhao Jia**, Wenhao Yu, Kaixin Ma, Tianqing Fang, Zhihan Zhang, Siru Ouyang, Hongming Zhang, Dong Yu, Meng Jiang. *Leopard: A Vision Language Model for Text-Rich Multi-Image Tasks.* **TMLR** 2025. [arXiv] [Code] [HuggingFace]
   In Leopard, we perform multimodal pre-training and instruction tuning, construct the Leopard-Instruct dataset, and design adaptive high-resolution image encoding for text-rich multi-image understanding. Leopard outperforms prior state-of-the-art models by an average of 9.61 points across five benchmarks. Leopard-Instruct has received over 200K downloads.
5. Zifeng Zhu*, **Mengzhao Jia***, Zhihan Zhang, Lang Li, Meng Jiang. *MultiChartQA: Benchmarking Vision-Language Models on Multi-Chart Problems.* **NAACL** 2025. [arXiv] [HuggingFace]
   MultiChartQA benchmarks MLLMs' ability to understand and integrate information across multiple charts. It requires models to connect evidence across charts through multi-hop, comparative, and sequential reasoning.

Peer-Reviewed Publications

6. **Jia**, Wei, Song, Sun, Zhang, Nie. *Query-Oriented Micro-Video Summarization.* IEEE **TPAMI**, 46 (6), 4174–4187, 2024.

7. **Jia**, Xie, Jing. *Debiasing Multimodal Sarcasm Detection with Contrastive Learning.* **AAAI** 2024.
8. **Jia**, Chen, Jing, Fu, Li. *Knowledge-Enhanced Memory Model for Emotional Support Conversation.* Machine Learning for Cognitive and Mental Health Workshop, **AAAI** 2024.
9. Yang, Guo, **Jia**, Su, Liu, Zhang, Jiang. *MMTutorBench: The First Multimodal Benchmark for AI Math Tutoring.* **ACL** 2026.
10. Liu, Dou, **Jia**, Tan, Zeng, Yuan, Jiang. *Protecting Privacy in Multimodal Large Language Models with MLLMU-Bench.* **NAACL** 2025.
11. Ouyang, Yu, Ma, Xiao, Zhang, **Jia**, et al. *RepoGraph: Enhancing AI Software Engineering with Repository-Level Code Graph.* **ICLR** 2025.
12. Zhang, Ge, Liang, Yu, Yu, **Jia**, Yu, Jiang. *Learn Beyond The Answer: Training Language Models with Reflection for Mathematical Reasoning.* **EMNLP** 2024.
13. Zhang, Lee, Fang, Yu, **Jia**, Jiang, Barbieri. *PLUG: Leveraging Pivot Language in Cross-Lingual Instruction Tuning.* **ACL** 2024.
14. Jing, Song, Ouyang, **Jia**, Nie. *Multi-source Semantic Graph-based Multimodal Sarcasm Explanation Generation.* **ACL** 2023.
15. Nie, **Jia**, Song, Wu, Cheng, Gu. *Multimodal Activation: Awakening Dialog Robots without Wake Words.* **SIGIR** 2021.

Service

Reviewer for ACL, ICLR, CVPR, AAAI